M. Tuna Alatan

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Education

Bilkent University, B.S. in Electrical and Electronics Engineering — Sept 2023 – June 2027

- GPA: 3.74/4.0

METU Foundation School, High School Degree — Sep 2019 – June 2023

- GPA: 93.9/100

Experience

Computer Vision Intern, ROMER – METU, Ankara — Jul 2025 – Aug 2025

- Designed a novel pipeline for **labeling 3D-LiDAR** data by utilizing **stereo-vision**, LiDAR-Gate, which earned the "**Best Presentation of the Internship**" award.
- Built an end-to-end perception pipeline using **FoundationStereo (depth)**, **YOLO & SegFormer (segmentation)**, **DBSCAN (clustering)**, and **PCA (primitive fitting)**.
- The GitHub repository is available at github.com/tuna-alatan/LiDAR-Gate.

Undergraduate Research Intern, METU, Ankara — Jun 2025 – Aug 2025

- Under the supervision of **Prof. Dr. Ahmet Oğuz Akyüz** conducted research on low-light image enhancement and color calibration using color conversion matrix.
- Conducted a comparative evaluation of low-light image enhancement methods**—Histogram Equalization, Plateau Equalization, and CLAHE—**focusing on visual quality and computational efficiency.
- Deployed **Zero-DCE on Raspberry Pi + Hailo AI Hat** for real-time low-light enhancement at the edge.
- Performed color calibration on low-light images captured from low-light-sensitive camera using **CCM and ColorChecker**.
- The GitHub repository is available at github.com/tuna-alatan/LowLightCalib.

Embedded Systems Intern, Artron – Hacettepe Teknokent, Ankara — Jul 2024 – Aug 2024

- Implemented basic image processing algorithms such as **filtering and bilinear interpolation** for **FPGAs** using **VHDL**.
- Designed image processing pipelines including **timing signal generation**, **test image synthesis**, and **block RAM-based buffering**.
- Simulated image data paths and timing behavior using **ModelSim**.
- The GitHub repository which contains the code and the internship report can be accessed through github.com/tuna-alatan/Intro-Image-Processing-VHDL.

Computer Vision Intern, OGAM – METU, Ankara — June 2021 – Aug 2021

- Worked with **Intel RealSense D435** depth camera using Python SDK for data acquisition and processing.

Awards

- **40% Merit Scholarship**, Ihsan Dogramaci Bilkent University, (2025-2026 Academic Year)
- **80% Merit Scholarship**, Ihsan Dogramaci Bilkent University, (2024-2025 Academic Year)
- **90% Merit Scholarship**, METU Foundation High School
- **2nd Place**, RIDERS Robotic League 2022 Season Final

Projects

Smart Camera System for Real-Time Wi-Fi Streaming and Target Tracking — GitHub Repository

- Developed a **smart camera system** that is mounted on two-axis servos which can be controlled via a joystick.
- The image data is streamed through **Wi-Fi** which can be accessed by devices that are connected to the local

Wi-Fi.

- A **tracking algorithm** consisting of integral image and **filtering** is embedded on the camera to track a black crosshair in a frame. The location of the crosshair in the frame is then mapped to an 8x8 LED matrix.
- Tools Used: KL25Z ARM Cortex M0+ MCU, ESP32-Cam Module, Embedded C.

Color Tracking Camera GitHub Repository

- Developed a **color-tracking** camera system that is controlled by an **FPGA**. A webcam and a servo motor is used to construct the rotating camera hardware. Python and VHDL is used for developing the software.
- Tools Used: BASYS3 FPGA, VHDL, Python.

Ray Tracing Engine

- Implemented a physically based ray tracer with **GGX BRDFs** and experimented with **CUDA** acceleration for performance optimization.
- Tools Used: C++, CUDA.

Technologies

Languages: C, C++, Python, VHDL, 8051 Assembly, CUDA, MATLAB.

Embedded & Hardware: FPGA (Vivado), ModelSim, LTspice, Proteus

Computer Vision & ML: OpenCV, YOLO, SegFormer, PyTorch, Numpy, FoundationStereo.

Creative & Visualization: Blender, Unreal Engine, Substance Painter, Adobe Photoshop, DaVinci Resolve.